

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 07



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Recap of Previous Lecture



Topic

Transitive relation

Topic

Complement of a relation

Topic

Inverse of a relation

Topic

Composite of two relations

Topics to be Covered



Topic

Reflexive, symmetric and transitive closure



Topic

Equivalence relation



Topic

Partition of a set





Topic : Reflexive Closure

* Let R is any relation on set A .

Reflexive closure of relation R is the smallest reflexive relation on set A that contains relation R

eg. let $A = \{1, 2, 3\}$

$R = \{(1, 1), (1, 2), (2, 1), (3, 2), (3, 3)\}$

$(1, 1) \in R, \checkmark$

$(3, 3) \in R, \checkmark$

but $(2, 2) \notin R \therefore R$ is not reflexive

Reflexive Closure of $R = \{(1, 1), (1, 2), (2, 1), (3, 2), (3, 3), (2, 2)\}$

all order pairs of relation R must be present

it is added to make it reflexive

Note $\rightarrow$ ① If given relation R is already a reflexive relation on set A , then reflexive closure of relation R will be relation R itself.

② Reflexive closure of $R = R \cup \Delta_A$
Where R is any relation on set A



Topic : Symmetric Closure

→ Let R is any relation on set A ,
Symmetric closure of relation R is the smallest
symmetric relation on set A that contains R

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)\}$

$\left\{ \begin{array}{l} R \text{ is not a} \\ \text{symmetric relation} \end{array} \right.$

Symmetric
Closure of $R = \{ \underbrace{(1,1), (1,2), (2,1), (2,2), (3,1), (3,2)}_{\text{all order pairs from } R}, \underbrace{(1,3), (2,3)}_{\text{added to make it symmetric}} \}$

Note: ① If R is symmetric relation on set A ,
then symmetric closure of relation R will be
relation R itself

② Symmetric closure of $R = R \cup R^{-1}$



Topic : Transitive Closure

- Let R is any relation on set A ,
Transitive closure of relation R is the
Smallest transitive relation on set A that
Contains relation R

eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

find transitive closure of R

eg. Let $A = \{1, 2, 3\}$

$R = \{(1, 1), (1, 2),$

ends with '2'

Order pairs starting with '2' is $(2, 2)$

$(1, 2) \& (2, 2) \Rightarrow \therefore (1, 2)$

already present

$(1, 3), (2, 2), (3, 1)\}$

ends with '3'

Order pairs starting with '3' is $(3, 1)$

$(1, 3) \& (3, 1) \Rightarrow \therefore (1, 1)$

already present

ends with '1'

Order pairs that starts with '1' are $(1, 1), (1, 2), (1, 3)$

$(3, 1) \& (1, 1) \Rightarrow \therefore (3, 1) \in R \checkmark$

$(3, 1) \& (1, 2) \Rightarrow \therefore (3, 2) \notin R$

$(3, 1) \& (1, 3) \Rightarrow \therefore (3, 3) \notin R$

$\therefore R$ is not transitive

✓ eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

find transitive closure of R

given Relation

$$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$$

$$R^* = \{(1,1), (1,2), (1,3), (2,2), (3,1) \mid (3,2), (3,3)\}$$

include all order pairs from R

$$R^{**} = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$$

include all order pairs from R^*

No new order pair is added
 $\therefore R^* = R^{**}$

★ Whenever the o/p of last two iteration are identical, then it is our transitive closure

$R^* = R^{**} \therefore R^*$ is transitive closure of R

eg. Let $A = \{1, 2, 3\}$

$$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

find transitive closure of R

- * R is already a transitive relation on set A ,
∴ Transitive closure of R is " R " itself

H.W.

eg.

Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

find transitive closure of R

eg. Let $A = \{a, b, c, d\}$

$$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$$

Find transitive closure of R ?

$$R = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c)\}$$

$$R^* = \{(a,d), (b,a), (b,c), (c,a), (c,d), (d,c) \mid (a,c), (b,d), (c,c), (d,a), (d,d)\}$$

include all order pairs of R

$$R^{**} = \{(a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d) \mid (a,a)\}$$

include all order pairs of R^*

$R^* \neq R^{**}$
 $\therefore R^*$ is not transitive closure of R .

$$R^{***} = \{(a,a), (a,c), (a,d), (b,a), (b,c), (b,d), (c,a), (c,c), (c,d), (d,a), (d,c), (d,d)\}$$

$R^{**} = R^{***}$, $\therefore R^{**}$ is transitive closure of R .

No new order pair is added

Note :- *

Warshall's algorithm can be used to obtain the transitive closure of the given relation

eg. Let $A = \{1, 2, 3\}$

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

find transitive closure of R using Warshall's algorithm

$R = \{(1,1), (1,2), (1,3), (2,2), (3,1)\}$

Step-1:- Obtain adjacency matrix w.r.t Relation R

Relation R

	I	II	III
I	1	1	1
II	0	1	0
III	1	0	0

Column No. (I, II, III)
Row number (I, II, III)

Step-2:-

	I	II	III
I	{1,3}	{1,2}	{1}
II	{1,2,3}	{2}	{1}
III	{(1,1), (1,2), (1,3), (3,1), (3,2), (3,3)}	{(1,2), (2,2)}	{(1,1)}

Column number (I, II, III)
Row number (I, II, III)
'1' entry in Column
'1' Entry in Row
Column x Row

Step-3: obtain

$R \cup \text{I}^{\text{st}} \text{ Column} \cup \text{II}^{\text{nd}} \text{ Column} \cup \text{III}^{\text{rd}} \text{ Column}$
 $\text{I}^{\text{st}} \times \text{Row} \quad \text{II}^{\text{nd}} \times \text{Row} \quad \text{III}^{\text{rd}} \times \text{Row}$

$= \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$

Repeat "Step-1", "Step-2" & "Step-3", until the o/p of "Step-3" is same for last two iteration

⑤ If we are asked to find the reflexive-transitive closure of relation R on set A .

- * Reflexive-transitive closure of ' R ' is the smallest reflexive as well as transitive relation on set A that contains " R ".



Topic : Equivalence relation

→ A relation R on set A is called an Equivalence relation if and only if relation R is

- and
- ① Reflexive
 - ② Symmetric
 - ③ Transitive

Equivalence Relation = Reflexive + Symmetric + Transitive

eg. Let $A = \{1, 2, 3\}$

$\Delta_A = R_1 = \{(1,1), (2,2), (3,3)\}$
Diagonal Relation on set A

Reflexive ✓

Symmetric ✓

Transitive ✓

} $\therefore R_1$ is an Equivalence Relation on set A

Δ_A is the smallest Equivalence Relation on set A

$R_2 = 'A \times A' = \begin{matrix} \text{Ref} \checkmark \\ \text{Symm} \checkmark \\ \text{Transitive} \checkmark \end{matrix}$

} $\therefore R_2$ is an Equivalence Relation

' $A \times A$ ' is the largest Equivalence Relation on set A

eg. let $A = \{1, 2, 3\}$

$R_3 = \{ \underbrace{(1,1), (2,2), (3,3)}_{\text{Reflexive}}, \underbrace{(1,2), (2,1)}_{\text{Symmetric}} \}$ $\therefore R_3$ is an Equivalence relation on set A

$\underbrace{\hspace{10em}}_{\text{Transitive}}$

$R_4 = \{ (1,1), (2,2), (3,3), (1,3), (3,1) \}$ R_4 is an Equivalence relation

$R_5 = \{ (1,1), (2,2), (3,3), (2,3), (3,2) \}$ R_5 is an Equivalence relation

eg. Let $A = \{1, 2, 3\}$

$$R_6 = \{ \underbrace{(1,1), (2,2), (3,3)}_{\text{Reflexive}}, \underbrace{(1,2), (2,1)}_{\text{Symmetric}}, \underbrace{(1,3), (3,1)}_{\text{Symmetric}} \}$$

$(1,2) \wedge (1,3) \Rightarrow \therefore (2,3) \notin R_6$
transitivity
 $\therefore$ not transitive

R_6 is $\begin{cases} \text{Reflexive} \checkmark \\ \text{Symmetric} \checkmark \end{cases}$

Transitive $\times \rightarrow \therefore$ Not an Equivalence relation

→ Check whether the statement is true or false.

"Every symmetric as well as transitive relation on set 'A', is also a reflexive relation on set A"

it is a "false" statement

eg let $A = \{1, 2, 3\}$

$R = \{ \}$
Empty relation
on set A

Symmetric ✓
Transitive ✓

} but not reflexive on set A

let $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (2,2), (1,2), (2,1)\}$

Symmetric ✓
Transitive ✓

But it is not reflexive



Topic : Equivalence Class

Let "R" is an Equivalence relation on set A.
Equivalence class of an element $x \in A$ w.r.t. Equivalence relation R can be denoted by $[x]$, and it is defined as

$$[x] = \{ y \mid x \underset{\text{'''}}{R} y \}$$

i.e., $[x] = \{ y \mid (x, y) \in R \}$

Equivalence class of element 'x' is the set of all elements that relates with 'x' w.r.t relation 'R'

eg: Let $A = \{1, 2, 3, 4, 5\}$

and R is an Equivalence relation on set A .

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$$

Find Equivalence class of every element of set A
w.r.t. Equivalence relation R .

Soln.

$$[1] = \{1, 2\}$$

Same
ie, $[1] = [2]$

$$[2] = \{2, 1\} = \{1, 2\}$$

$$[3] = \{3\}$$

$$[4] = \{4, 5\}$$

Same.
ie, $[4] = [5]$

$$[5] = \{5, 4\} = \{4, 5\}$$

Home Work:.

- ① Read about Partitions of a set.



2 mins Summary



✓ **Topic**

Reflexive, symmetric and transitive closure

✓ **Topic**

Equivalence relation

Topic

Partition of a set



THANK - YOU